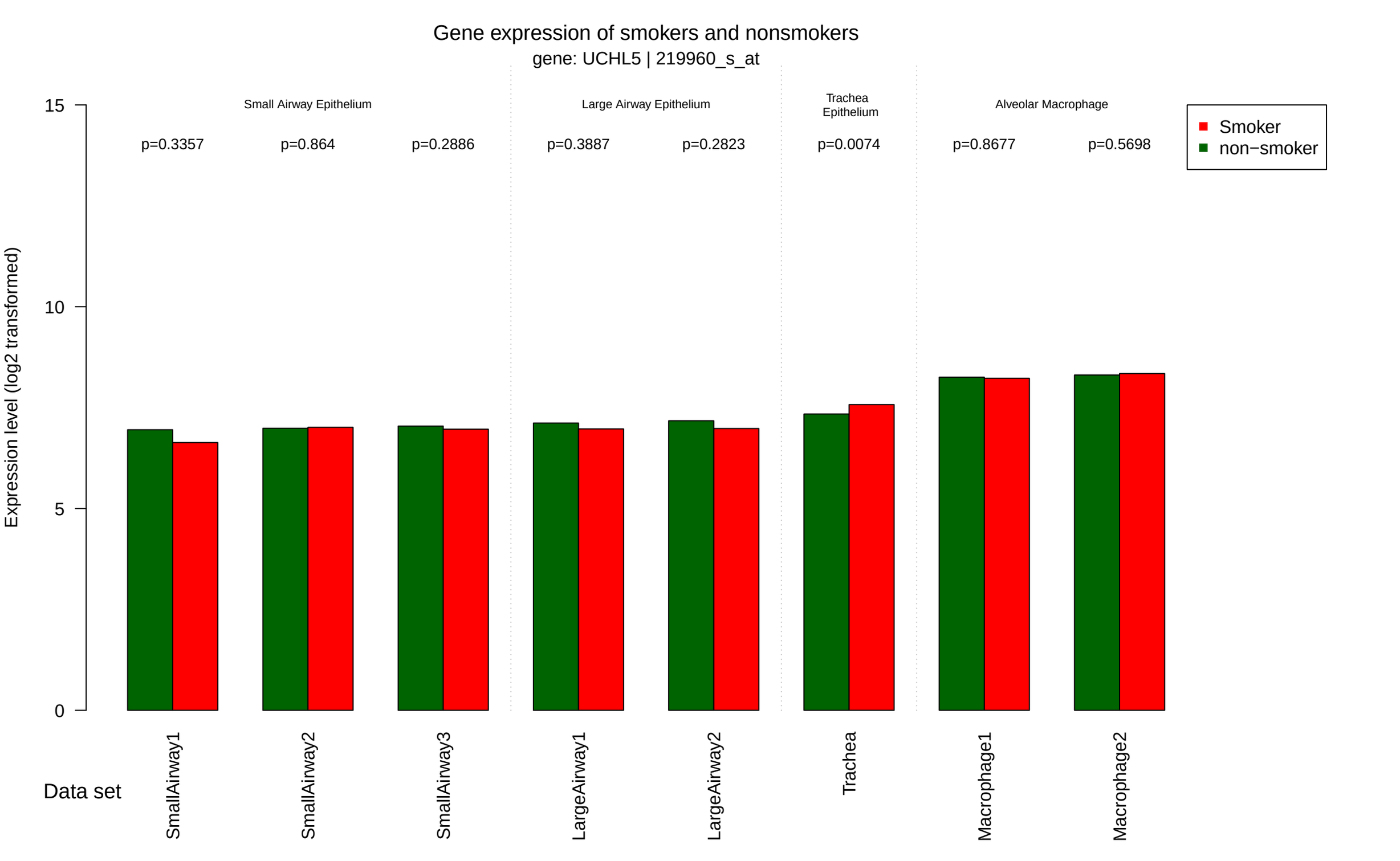
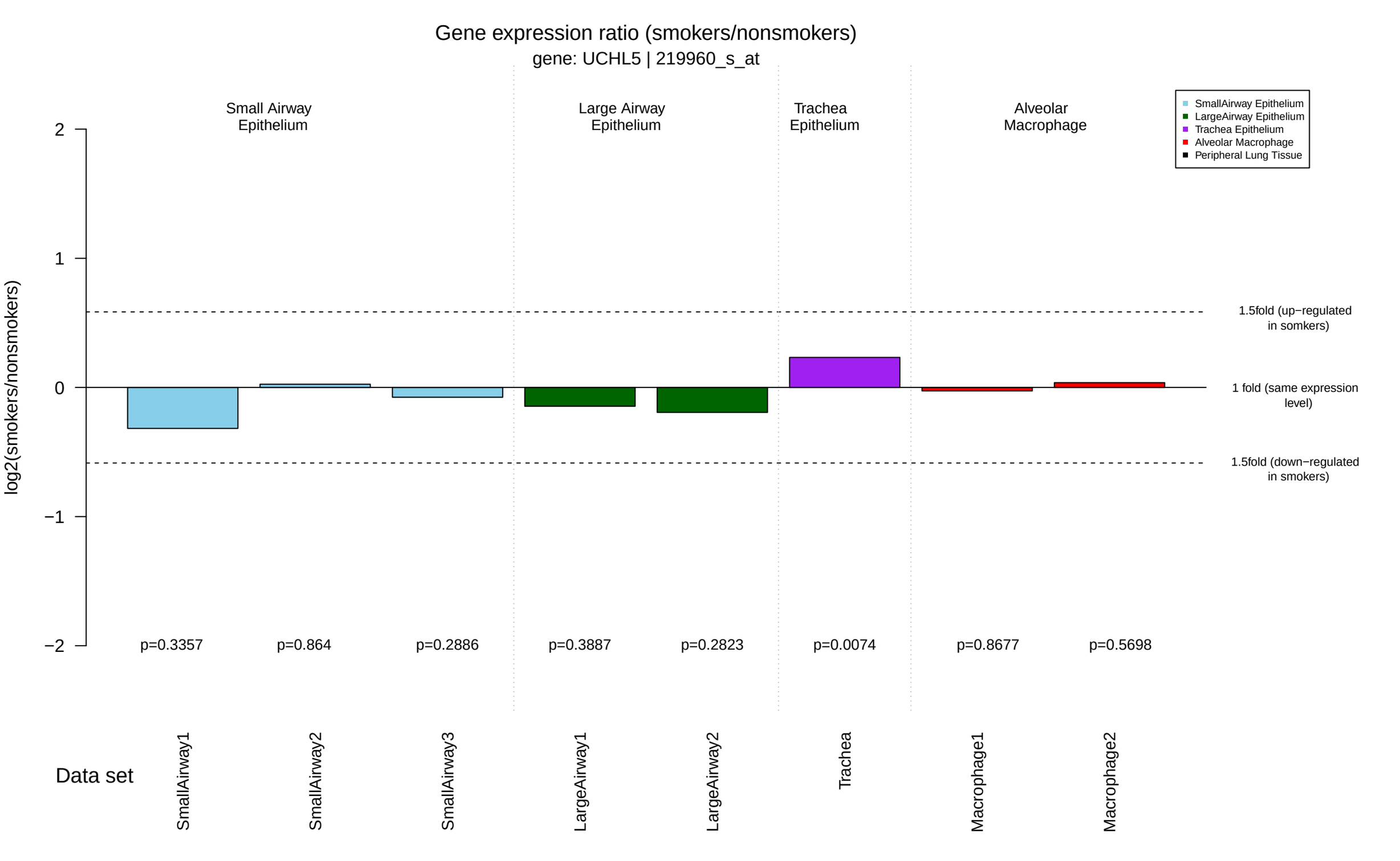
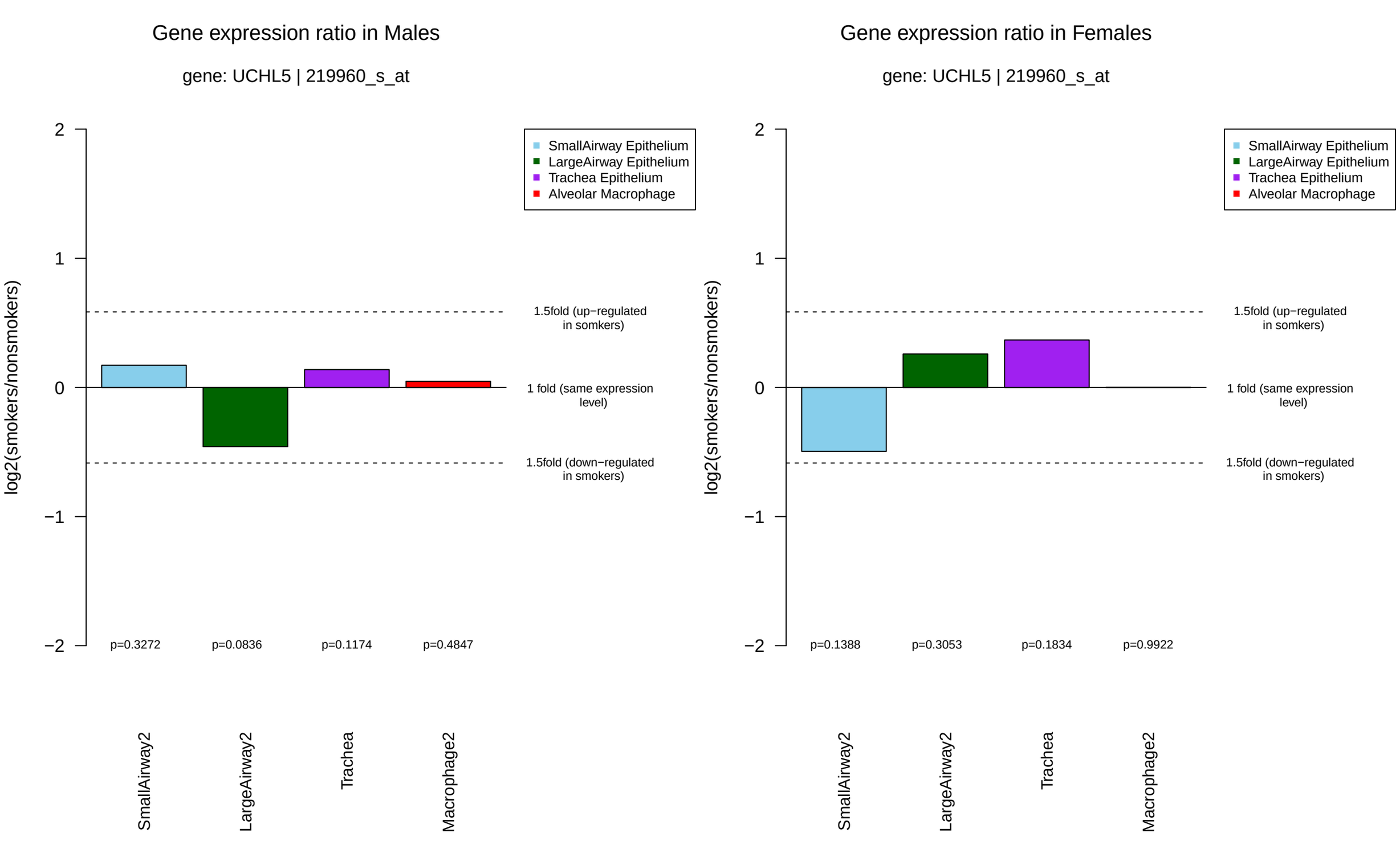
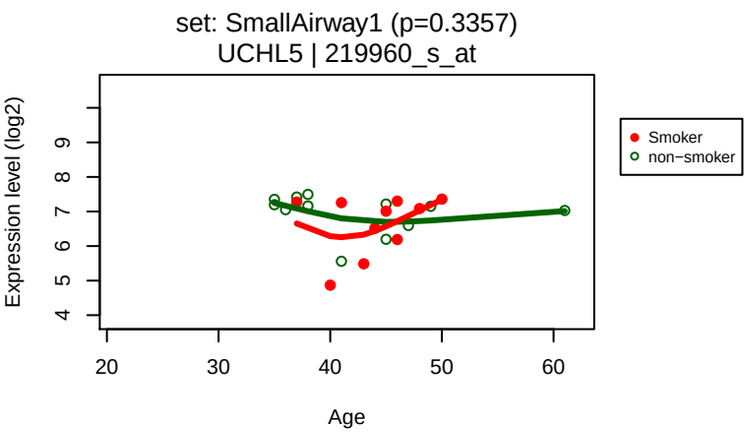
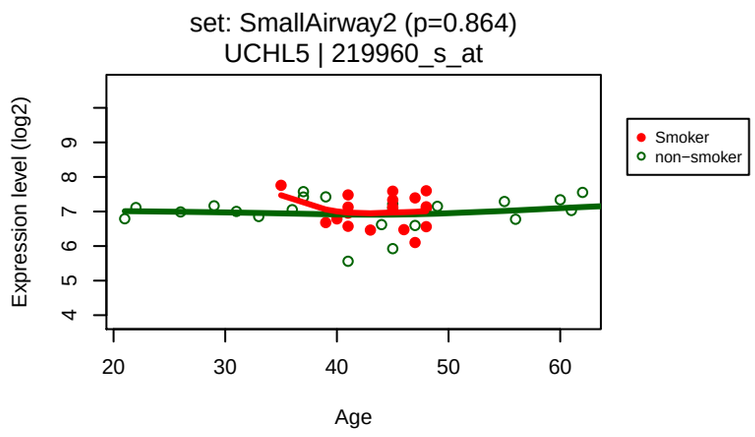
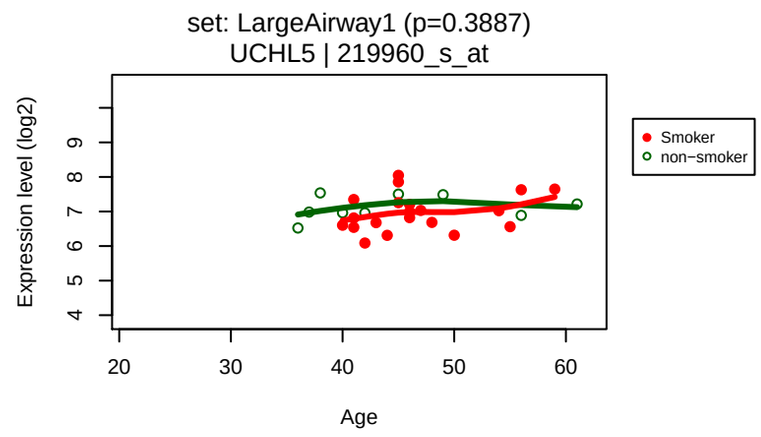

Gene expression of smokers and nonsmokers
gene: UCHL5 | 219960_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3357
p=0.864
p=0.2886
p=0.3887
p=0.2823
p=0.0074
p=0.8677
p=0.5698
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

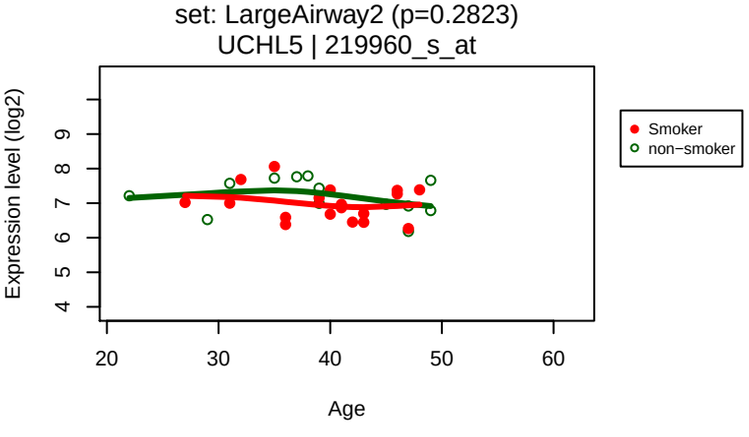

Gene expression ratio (smokers/nonsmokers)
gene: UCHL5 | 219960_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3357
p=0.864
p=0.2886
p=0.3887
p=0.2823
p=0.0074
p=0.8677
p=0.5698
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

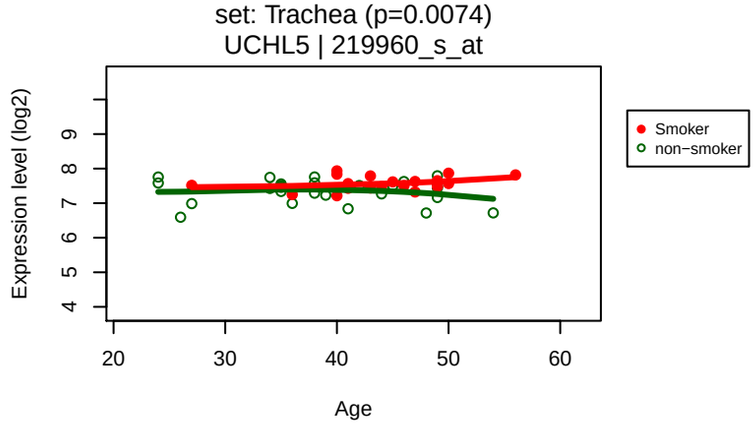

Gene expression ratio in Males
gene: UCHL5 | 219960_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3272
p=0.0836
p=0.1174
p=0.4847
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: UCHL5 | 219960_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1388
p=0.3053
p=0.1834
p=0.9922
SmallAirway2
LargeAirway2
Trachea
Macrophage2

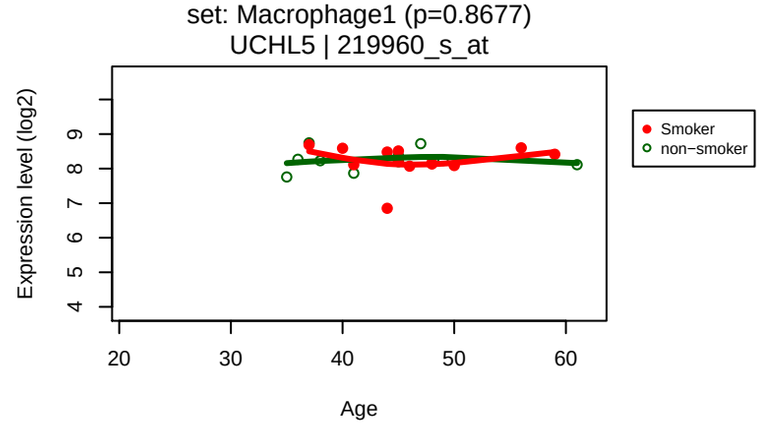

set: SmallAirway1 (p=0.3357)
UCHL5 | 219960_s_at
Expression level (log2)
Age
Smoker
non−smoker

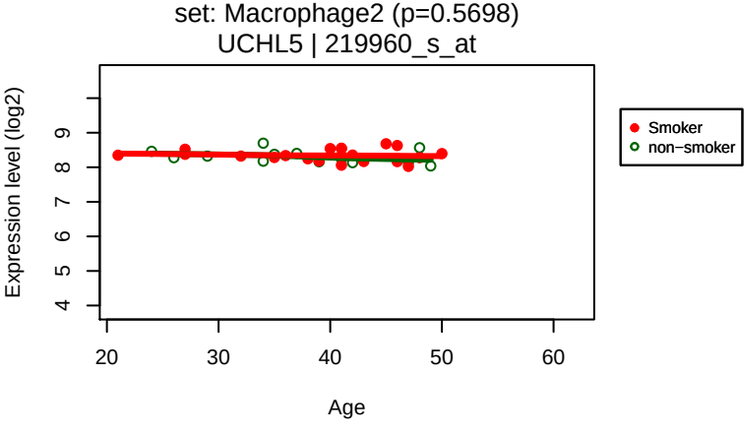

set: SmallAirway2 (p=0.864)
UCHL5 | 219960_s_at
Expression level (log2)
Age
Smoker
non−smoker

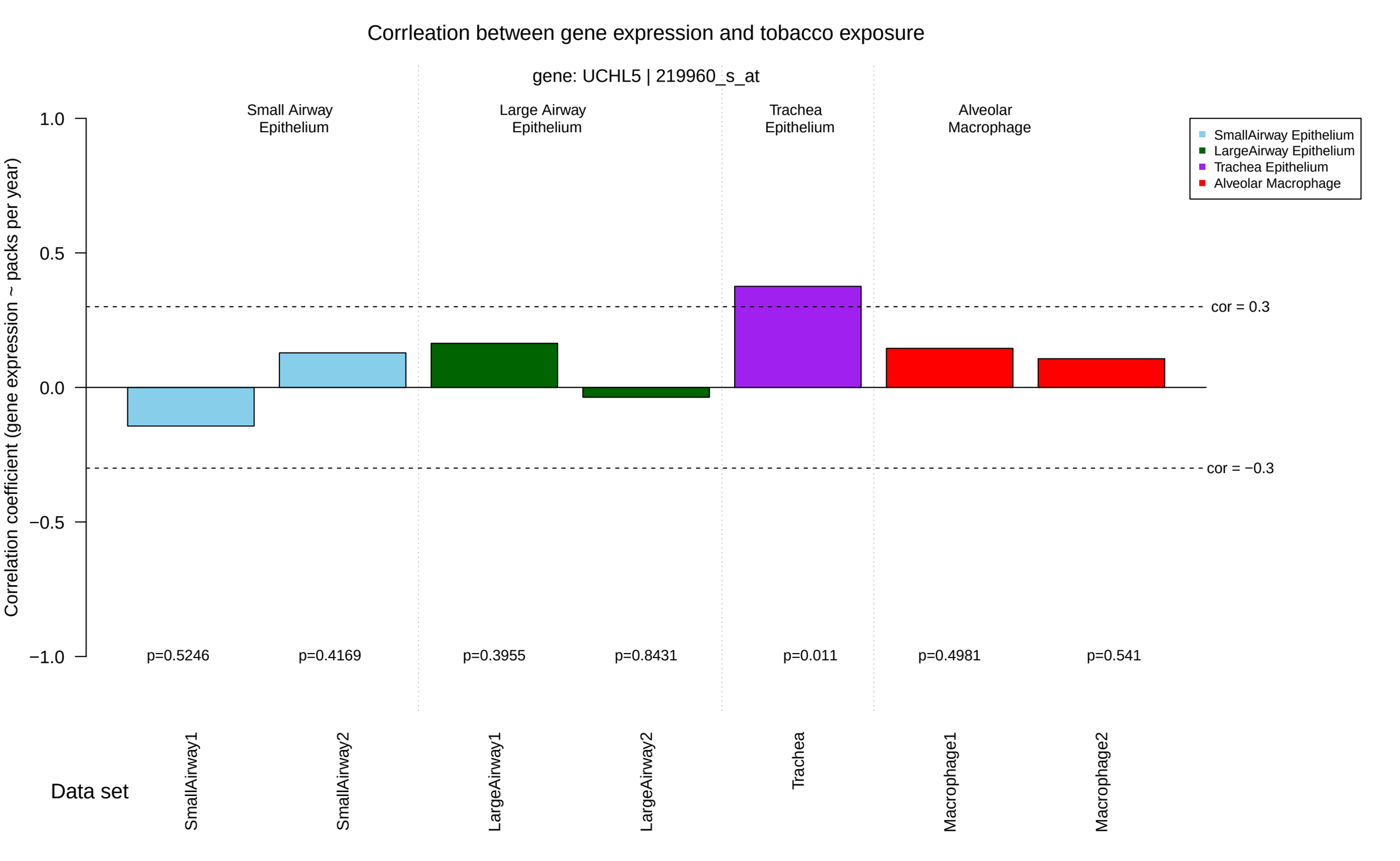

set: LargeAirway1 (p=0.3887)
UCHL5 | 219960_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2823)
UCHL5 | 219960_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0074)
UCHL5 | 219960_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8677)
UCHL5 | 219960_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5698)
UCHL5 | 219960_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: UCHL5 | 219960_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5246
p=0.4169
p=0.3955
p=0.8431
p=0.011
p=0.4981
p=0.541
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2